

Subject Description Form

Subject Code	LSGI4346
Subject Title	Deformation Monitoring and Tunnel Alignment: Advanced Surveying Practices
Credit Value	4
Level	4
Pre-requisite	LSGI3781 Engineering Surveying
Objectives	The aims of this subject are: • To equip students with advanced technical skills to design, implement, and manage modern deformation monitoring schemes and precise tunnel alignment surveys. • To provide a thorough understanding of the principles underlying advanced surveying technologies—including satellite positioning, InSAR, laser scanning, and gyrotheodolites—for monitoring and control. • To develop proficiency in processing, analysing, and interpreting deformation and alignment data using geophysical modelling, statistical methods, and free network adjustment • To foster critical and creative thinking through the evaluation of surveying strategies, data quality, and real-world engineering challenges; and • To cultivate professional attitudes and collaborative problem-solving skills through group projects, practical field exercises, and technical discussions.
Intended Learning Outcomes	At the end of this subject students who gain a pass should be able to: a. Design and implement advanced deformation monitoring schemes for various objects such as ground, slopes, and man-made structures, selecting appropriate geodetic and geotechnical methods (L4) b. Process and rigorously analyse data from deformation surveys, performing statistical tests and free network adjustments to ascertain significance and reliability. (L4) c. Model and interpret deformations by integrating survey measurements with geophysical and statistical models to infer causal mechanisms and predict future behaviour (L3) d. Plan and execute precise surveys for tunnel alignment control, including the transfer of control underground and the use of specialized instrumentation like gyrotheodolites (L4) e. Critically evaluate the performance and limitations of different surveying technologies and methodologies in the context of deformation monitoring and tunnel engineering projects (L4)
Subject Synopsis/ Indicative Syllabus	A. Advanced Deformation Monitoring and Analysis • Fundamentals & Scheme Design: Concepts of deformation; geotechnical and structural behaviour; principles of designing robust and reliable deformation monitoring schemes. • Advanced Monitoring Technologies:

	○ Geodetic Methods: High-precision GNSS/GPS for continuous monitoring. ○ Remote Sensing: Satellite Radar Interferometry (InSAR) for wide-area deformation mapping. ○ Terrestrial & Mobile Sensing: Laser scanning (LiDAR) and photogrammetry for 3D displacement analysis. • Data Processing & Robust Analysis: Advanced least-squares adjustment, including Free Network Adjustment; statistical methods for detecting and quantifying significant movements; data filtering and trend analysis. • Deformation Modelling & Interpretation: Modelling deformation using geophysical and geostatistical models; integrating sensor data to interpret causes and predict future behaviour. B. Precision Tunnel Alignment and Control Surveys • Tunnelling & Surveying Integration: Overview of tunnelling methods and their specific survey requirements; the role of surveying throughout the project lifecycle. • Advanced Underground Control: ○ Precise transfer of azimuth and coordinates from surface to underground. ○ Operation, application, and precision limits of gyrotheodolites and inertial measurement systems. • Guidance and Verification Surveys: Guidance for Tunnel Boring Machines (TBMs) and conventional tunnelling; monitoring of interim and final tunnel alignment. • Break-through and As-Built Surveys: Design and execution of high-precision break-through surveys; verification of final alignment through as-built surveys and volumetric analysis.
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Teaching/Learning Methodology	The teaching and learning philosophy for this subject is based on a blended and experiential approach, designed to bridge advanced theoretical concepts with professional practice. • Blended Learning Delivery: A combination of interactive lectures, online resources, and hands-on practical sessions is used. Core theoretical materials are provided via the PolyU Blackboard for preparatory study, freeing up class time for advanced discussion, problem-solving, and application. • Technology-Enhanced Learning: Students will engage with industry-standard software for data processing (e.g., least-squares adjustment, statistical analysis) and be introduced to the operation and data outputs of advanced instruments (e.g., GNSS, laser scanners, gyrotheodolites) through simulations and virtual site tours where applicable. • Project-Based and Practical Learning: Students' technical skills and professional judgment will be developed through a series of integrated practical projects. These projects simulate real-world scenarios, requiring students to design a monitoring scheme, process and analyse collected data, and interpret the results in a geotechnical or construction context. • Development of Higher-Order Thinking Skills: Critical and creative thinking is fostered through case study analysis of real project challenges, such as the interpretation of complex deformation patterns or troubleshooting alignment errors in tunnels. Problem-based learning is embedded in tutorials and assignments. • Cultivation of Professional Practice: Collaborative attitudes and team-working skills, essential for the surveying profession, are cultivated through group projects and lab sessions. Students will practice effective communication through technical reporting and presentations of their findings, mirroring professional consultancy practice.																																																				
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>1. Technical Report: Deformation Monitoring Project</td><td>25</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td></tr> <tr> <td>2. Phase Tests (2)</td><td>40</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr> <tr> <td>3. Practical Portfolio: Tunnel Alignment & Control</td><td>25</td><td></td><td></td><td></td><td>√</td><td>√</td></tr> <tr> <td>4. Professional Engagement & Collaboration</td><td>10</td><td>√</td><td></td><td></td><td>√</td><td>√</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="4"></td><td></td></tr> </tbody> </table>						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Technical Report: Deformation Monitoring Project	25	√	√	√		√	2. Phase Tests (2)	40	√	√	√	√	√	3. Practical Portfolio: Tunnel Alignment & Control	25				√	√	4. Professional Engagement & Collaboration	10	√			√	√	Total	100 %					
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	Description of Assessment Methods: • 1. Technical Report: Deformation Monitoring Project This report assesses the ability to design a monitoring scheme (a), process and rigorously analyze the data (b), model and interpret the results to draw meaningful conclusions (c), and critically evaluate the methodology and results (e). • 2. Phase Tests (2) These tests comprehensively evaluate students' understanding of theoretical principles and their ability to apply knowledge across all topics. They include problems designed to assess critical thinking and the evaluation (e) of different technological approaches. • 3. Practical Portfolio: Tunnel Alignment & Control This portfolio compiles evidence from field exercises, including data sheets, calculations, and a reflective summary. It directly assesses the competency to plan and execute (d) precise surveys and the ability to critically evaluate (e) the quality and precision of the achieved results. • 4. Professional Engagement & Collaboration This component assesses professional skills through continuous observation during group projects and practicals. It evaluates the initiative and collaborative attitude in designing and implementing schemes (a), executing practical work (d), and engaging in critical discussion and peer review (e)	
Student Study Effort Expected	Class contact:	
	▪ Lecture/tutorials	26 Hrs.
	▪ Practical work	39 Hrs.
	▪ Site visits	6 Hrs.
	Other student study effort:	
	▪ Guided Reading & Pre/Post-Class Preparation (incl. online materials)	24 Hrs.
	▪ Deformation Monitoring Project (Research, Data Analysis, Report Writing)	25 Hrs.
	▪ Practical Portfolio Compilation (Data Processing, Calculations, Reflection)	15 Hrs.
	▪ Exam/Test Preparation	8 Hours
	Total student study effort:	143 Hrs.

Reading List and References	Core Textbooks: 1. Scaioni, M., et al. (Eds.) (2015). <i>Modern Technologies for Landslide Investigation and Prediction</i>. Springer. <i>Relevance: Provides a modern, comprehensive look at geodetic and remote sensing methods for deformation monitoring, directly supporting the advanced syllabus.</i> 2. Walker, J., & Awange, J. (2020). <i>Surveying for Civil and Mine Engineers: Theory, Workshops, and Practicals</i> (2nd ed.). Springer. <i>Relevance: Excellent for both deformation monitoring and tunnel surveying, with a strong emphasis on practical application and modern techniques like GNSS.</i> 3. Bai, Y. (2019). <i>Underground Engineering: Planning, Design, Construction and Operation of the Underground Space</i>. Elsevier. <i>Relevance: Offers a broad engineering context for tunnel projects, which is essential for understanding the role and requirements of alignment control surveys.</i> Key References (Deformation Monitoring): 4. Ketelaar, G. (V.B.H.) (2009). <i>Satellite Radar Interferometry: Subsidence Monitoring Techniques</i>. Springer. <i>Relevance: A foundational text specifically on InSAR, a key technology mentioned in the syllabus.</i> 5. Caspary, W. (1988). <i>Concepts of Network and Deformation Analysis</i>. University of New South Wales. <i>Relevance: Remains a crucial theoretical text for the principles of free network adjustment and deformation analysis.</i> 6. Hanssen, R. F. (2001). <i>Radar Interferometry: Data Interpretation and Error Analysis</i>. Springer. <i>Relevance: A standard reference for understanding the principles and challenges of InSAR data processing.</i> Key References (Tunnel Surveying & Engineering): 7. Richardus, P. (1984). <i>Project Surveying</i> (2nd ed.). A.A. Balkema. <i>Relevance: A classic text that covers fundamental principles of tunnel surveying, including gyrotheodolite use.</i> 8. Uren, J., & Price, W.F. (2010). <i>Surveying for Engineers</i> (5th ed.). Palgrave Macmillan. <i>Relevance: Contains accessible chapters on setting out and underground surveying, useful for foundational knowledge.</i>
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	9. Hofmann-Wellenhof, B., Lichtenegger, H., & Wasle, E. (2008). <i>GNSS – Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and more</i>. Springer. <i>Relevance: Provides the essential theoretical background for high-precision satellite positioning used in both deformation and tunnel control networks.</i> 10. Schofield, W. (2007) Engineering Surveying, Butterworth-Heinemann. 11. Kavanagh, B.F. (2000) Surveying with Construction Applications, Pearson Prentice Hall. 12. Leick, A. (2004) Satellite Surveying, Wiley. 13. Uren, J., & Price, W.F. (2010) Surveying for Engineers, Macmillan. 14. William I. and F. Maclellan, 2006, Surveying for Construction, 5th Edition, 2006, McGraw-Hill.
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